

- for any time-invariant random variables  $Z_{i,t}$  such that  $Z_{i,t} = Z_i$ , double-demeaning annihilates it:  
 $\ddot{Z}_{i,t} = 0$ .

*Proof.* The zero-mean and zero-sum properties hold by definition of double-demeaning:

$$E[\ddot{X}_{i,t}] = E\left[X_{i,t} - E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T X_{i,s} + \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}]\right] \quad (33)$$

$$= E[X_{i,t}] - E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}] + \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}] = 0 \quad (34)$$

and

$$\sum_{t=0}^T \ddot{X}_{i,t} = \sum_{t=0}^T X_{i,t} - \sum_{t=0}^T E[X_{i,t}] - \sum_{t=0}^T \frac{1}{T+1} \sum_{s=0}^T X_{i,s} + \sum_{t=0}^T \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}] \quad (35)$$

$$= \sum_{t=0}^T X_{i,t} - \sum_{t=0}^T E[X_{i,t}] - \sum_{s=0}^T X_{i,s} + \sum_{s=0}^T E[X_{i,s}] = 0 \quad (36)$$

For the idempotent property, first note that by definition of double-demeaning, rearranging terms we can write  $E[\ddot{X}_{i,t} \ddot{Z}_{i,t}]$  as

$$= E\left[\left(X_{i,t} - \frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \ddot{Z}_{i,t}\right] - E\left[\left(E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}]\right) \ddot{Z}_{i,t}\right] \quad (37)$$

$$= E\left[\left(X_{i,t} - \frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \ddot{Z}_{i,t}\right] - \left(E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}]\right) \cdot E[\ddot{Z}_{i,t}] \quad (38)$$

By the zero-mean property  $E[\ddot{Z}_{i,t}] = 0$ , the second term of the above expression is zero. Summing the first term over  $t$ , we have

$$\sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}] - \sum_{t=0}^T E\left[\left(\frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \ddot{Z}_{i,t}\right] = \sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}] - E\left[\left(\frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \sum_{t=0}^T \ddot{Z}_{i,t}\right] = \sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}] \quad (39)$$

where the last equality follows from the zero-sum property  $\sum_{t=0}^T \ddot{Z}_{i,t} = 0$ . This proves  $\sum_{t=0}^T E[\ddot{X}_{i,t} \ddot{Z}_{i,t}] = \sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}]$ . Similarly we can show  $\sum_{t=0}^T E[\ddot{X}_{i,t} \ddot{Z}_{i,t}] = \sum_{t=0}^T E[\ddot{X}_{i,t} Z_{i,t}]$ .

The annihilating property of double-meaning holds by definition for any time-invariant random variable. Specifically, plugging in  $Z_{i,t} = Z_i$  we have

$$\ddot{Z}_{i,t} = Z_{i,t} - E[Z_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T Z_{i,s} + \frac{1}{T+1} \sum_{s=0}^T E[Z_{i,s}] \quad (40)$$

$$= Z_i - E[Z_i] - Z_i + E[Z_i] = 0 \quad (41)$$

□

## A.1 Expression of the weight

In this section, we provide detailed definitions for the three elements in the expression of the weight  $\omega_{e,\ell}^\ell = \sigma'_{e,\cdot} \mathbf{\Delta}_{e+\ell'} A_\ell^{-1}$ . Recall  $\omega_{e,\ell}^\ell$  is the weight associated with  $CATT_{e,\ell'}$  in the linear combination underlying  $\mu_\ell$ :

$$\begin{aligned}
\omega_{e,\ell'}^\ell &= \mathbf{e}_\ell^\top \underbrace{\left( \sum_{t=0}^T E[\mathbf{\ddot{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1}}_{:= (A_\ell^{-1})^\top} E[\mathbf{\ddot{D}}_{i,t} D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\}] \\
&= (A_\ell^{-1})^\top \sum_{t=0}^T Cov(\mathbf{D}_{i,t} - \frac{1}{T+1} \sum_{t=0}^T \mathbf{D}_{i,t}, D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\} - \frac{1}{T+1} \sum_{t=0}^T D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\}) \\
&= (A_\ell^{-1})^\top Pr\{E_i = e\} \cdot \left( (E[\mathbf{D}_{i,e+\ell'} - \frac{1}{T+1} \sum_{t=0}^T \mathbf{D}_{i,e+\ell'} | E_i = e]) - (E[\mathbf{D}_{i,e+\ell'}] - E[\frac{1}{T+1} \sum_{t=0}^T \mathbf{D}_{i,e+\ell'}]) \right) \\
&= (A_\ell^{-1})^\top \sum_{e'} E[\mathbf{D}_{i,e+\ell'} - \frac{1}{T+1} \sum_t \mathbf{D}_{i,t} | E_i = e'] \cdot Cov(\mathbf{1}\{E_i = e\}, \mathbf{1}\{E_i = e'\}) \\
&:= (\sigma_{e,\cdot})^\top \mathbf{\Delta}_{e+\ell'} A_\ell^{-1}
\end{aligned}$$

We note that

- The first term is the vector  $\sigma_{e,\cdot} = (\sigma_{e,e'})_{e' \in \text{supp}(E_i)}$  of the covariance between cohort indicators. Each entry  $\sigma_{e,e'} = Cov(\mathbf{1}\{E_i = e\}, \mathbf{1}\{E_i = e'\})$  is the covariance between being in cohort  $e$  and being in cohort  $e'$ .
- The second term is the matrix  $\mathbf{\Delta}_{e+\ell'}$  of demeaned relative time indicators. The rows of the matrix are indexed by cohorts  $e' \in \text{supp}(E_i)$  and the columns of the matrix are indexed by relative time indicators included in the specification (7). The entry that corresponds to cohort  $e'$  and relative time indicator  $D_{i,t}^\ell$  is  $E[D_{i,e+\ell'}^\ell - \frac{1}{T+1} \sum_t D_{i,t}^\ell | E_i = e']$ . Note that the sum  $\sum_t D_{i,t}^\ell$  is equal to  $\mathbf{1}\{E_i \in \mathcal{I}_\ell\}$  the indicator for whether this individual ever experience the relative time  $\ell$ .
- The third term is the row vector of  $A^{-1}$  that corresponds to the relative time indicator  $D_{i,t}^\ell$ . Here  $A$  is the covariance matrix of demeaned relative time indicators included in the specification (7). Specifically, the entry that corresponds to the covariance between demeaned  $D_{i,t}^\ell$  and  $D_{i,t}^{\ell'}$  is

$$A = \sum_t Cov\left(D_{i,t}^\ell - \frac{1}{T+1} \sum_t D_{i,t}^\ell, D_{i,t}^{\ell'} - \frac{1}{T+1} \sum_t D_{i,t}^{\ell'}\right).$$

Plugging in  $\sum_t D_{i,t}^\ell = \mathbf{1}\{E_i \in \mathcal{I}_\ell\}$  gives the expression in the main text.

## B Proofs of decomposition results

### Proof of Proposition 1

*Proof.* Collect the relative time indicators in a column vector  $\mathbf{D}_{i,t} = (\mathbf{1}\{t - E_i \in g\})_{g \in \mathcal{G}}^\top$ . Collect their corresponding coefficients in a column vector  $\mu = (\mu_g)_{g \in \mathcal{G}}^\top$ . Partialling out the unit and time fixed effects, Regression (4) is

$$\ddot{Y}_{i,t} = \mu^\top \ddot{\mathbf{D}}_{i,t} + v_{i,t} \quad (42)$$

where  $\ddot{X}_{i,t}$  is time- and cross-sectional demeaned version of  $X_{i,t}$  as defined in Definition 3. By the idempotent property of Lemma 1

$$\mu_g = \mathbf{e}_g^\top \left( \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,t}] \quad (43)$$

where  $\mathbf{e}_g$  is a column vector with one in the entry corresponding to the entry of  $\mathbf{1}\{t - E_i \in g\}$  in  $\mathbf{D}_{i,t}$ , and zero otherwise.

To further develop the expression for the population regression coefficients  $\mu_g$ , we note that by Lemma 1, we have  $\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,0}^\infty] = \sum_{t=0}^T E[\mathbf{D}_{i,t} \ddot{Y}_{i,0}^\infty] = 0$  since  $Y_{i,0}^\infty$  is time-invariant, and  $E[\ddot{\mathbf{D}}_{i,t}] = 0$  by the zero-mean property.

$$\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,t}] = \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,t}] - \underbrace{\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,0}^\infty]}_{=0} = \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} (Y_{i,t} - Y_{i,0}^\infty)] = \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} E[Y_{i,t} - Y_{i,0}^\infty | E_i]] \quad (44)$$

$$= \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} E[Y_{i,t} - Y_{i,0}^\infty | E_i]] - \underbrace{\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t}] E[Y_{i,t}^\infty - Y_{i,0}^\infty]}_{=0} \quad (45)$$

$$= \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} (E[Y_{i,t} - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty])]. \quad (46)$$

We can abbreviate the term in the parentheses as  $f(E_i, t)$  to emphasize it is a function of  $E_i$  and  $t$ . Since  $E_i$  and  $t$  take on discrete values, we can write

$$E[Y_{i,t} - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty] =: f(E_i, t) = \sum_{e=0}^{\infty} f(e, t) \cdot \mathbf{1}\{E_i = e\} \quad (47)$$

$$= \sum_{e=0}^{\infty} \sum_{\ell=-T}^T f(e, e + \ell) \cdot \mathbf{1}\{t - e = \ell\} \cdot \mathbf{1}\{E_i = e\} \quad (48)$$

$$= \sum_{\ell=-T}^T \sum_{e=0}^{\infty} D_{i,t}^\ell \mathbf{1}\{E_i = e\} \cdot f(e, e + \ell). \quad (49)$$

Even though we sum over  $e \in \{0, \dots, \infty\}$ , if there are no never-treated units (or no units in a particular cohort

$e'$ ), then we have no units take on values  $E_i = \infty$  (or  $E_i = e'$ ). The range of the summation is therefore still sensible. Note that we replace  $\mathbf{1}\{t - e = \ell\} \cdot \mathbf{1}\{E_i = e\}$  with  $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$  because

$$D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\} = \mathbf{1}\{t - E_i = \ell\} \cdot \mathbf{1}\{E_i = e\} = \mathbf{1}\{t - e = \ell\} \cdot \mathbf{1}\{E_i = e\}. \quad (50)$$

Using the above expression, the coefficient  $\mu_g$  can be written as

$$\mu_g = \mathbf{e}_g^\top \left( \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T \sum_{\ell=-T}^T \sum_{e=0}^\infty E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}] f(e, e + \ell) \quad (51)$$

$$= \sum_{t=0}^T \sum_{l \in g} \sum_{e=0}^\infty \underbrace{\mathbf{e}_g^\top \left( \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}]}_{=\omega_{e,\ell}^g} f(e, e + \ell) \quad (52)$$

$$+ \sum_{t=0}^T \sum_{g' \neq g} \sum_{\ell \in g'} \sum_{e=0}^\infty \underbrace{\mathbf{e}_g^\top \left( \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}]}_{=\omega_{e,\ell}^g} f(e, e + \ell) \quad (53)$$

$$+ \sum_{t=0}^T \sum_{\ell \in g^{excl}} \sum_{e=0}^\infty \underbrace{\mathbf{e}_g^\top \left( \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}]}_{=\omega_{e,\ell}^g} f(e, e + \ell). \quad (54)$$

The superscript  $g$  in  $\omega_{e,\ell}^g$  indexes  $\mu_g$ . The subscript  $\ell$  in  $\omega_{e,\ell}^g$  indexes  $f(e, e + \ell)$ . The expression above the braces makes it clear that the weight  $\omega_{e,\ell}^g$  is equal to the population regression coefficient on  $\mathbf{1}\{t - E_i \in g\}$  from regressing  $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$  on all the bin indicators i.e.  $\{\mathbf{1}\{t - E_i \in g\}\}_{g \in \mathcal{G}}$  and two-way fixed effects included in (4). This proves Proposition 1. We next show several properties of these weights in the expression of  $\mu_g$ :

1. For relative times of  $\mu_g$ 's own bin i.e.  $\ell \in g$ , the weights sum to one  $\sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = 1$ .
2. For relative times belonging to some other bin i.e.  $\ell \in g'$  and  $g' \neq g$ , the weights sum to zero  $\sum_{\ell \in g'} \sum_e \omega_{e,\ell}^g = 0$  for each bin  $g'$ .
3. For relative times not contained in  $\mathcal{G}$  i.e.  $\ell \in g^{excl}$ , the weights sum to negative one  $\sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g = -1$ .
4. If there are never-treated units i.e.  $\infty \in \text{supp}(E_i)$ , we have  $\omega_{\infty,\ell}^g = 0$  for all  $g$  and  $\ell$ .